Assignment module 6: Network Security, Maintenance, and Troubleshooting Procedures

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Section 1: Multiple Choice

Here are the correct answers with short explanations for your assignment:

1. **What is the primary purpose of a firewall?**

   Answer: b) Filtering and controlling network traffic
   A firewall monitors incoming and outgoing traffic and allows or blocks data based on security rules.

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2. What type of attack floods a network with excessive traffic?
Answer: a) Denial of Service (DoS)
A DoS attack overloads the system or network, making it unavailable to users.

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3. Which encryption protocol is commonly used for wireless security?
Answer: b) WPA (Wi-Fi Protected Access)
WPA (and WPA2/WPA3) is widely used to secure Wi-Fi networks. WEP is outdated and insecure.

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4. What is the purpose of a VPN?

Answer:

A VPN (Virtual Private Network) creates a secure, encrypted connection over the internet. It protects data privacy, hides the user's IP address, and allows safe remote access to networks.

Section 2: True or false

True or False: Patch management is the process of regularly updating software and firmware to address security vulnerabilities and improve system performance.

Answer: True

Patch management means regularly updating software and firmware to fix security vulnerabilities and improve performance.

True or False: A network administrator should perform regular backups of critical data to prevent data loss in the event of hardware failures, disasters, or security breaches.

Answer: True

Network administrators should always take backups to protect data from hardware failure, disasters, or cyber attacks.

True or False: Traceroute is a network diagnostic tool used to identify the route and measure the latency of data packets between a source and destination device.

Answer: True

Traceroute is used to track the path (route) and measure delay (latency) of packets from source to destination.

Section 3: Short

8. Describe the steps involved in conducting a network vulnerability Assignment.

> Planning & Scope Definition

Decide what systems, networks, and devices will be tested.

> Information Gathering (Reconnaissance)

Collect details like IP addresses, network structure, and services running.

> Scanning

Use tools to scan the network for open ports, services, and possible weaknesses.

> Vulnerability Identification

Identify security flaws such as outdated software, weak passwords, or misconfigurations.

>Risk Analysis

Evaluate how serious each vulnerability is and its potential impact.

> Reporting

.Prepare a report with all findings, risks, and recommended solutions.

- **Remediation**
  Fix the vulnerabilities by applying patches, updating systems, or changing configurations.
- **Verification (Re-testing)**
  Re-scan the network to ensure all issues have been resolved.

Section 4: Practical Application

9. Demonstrate how to troubleshoot network connectivity issues using the ping command.

Ping is a basic network tool used to check connectivity between devices.

Steps:

1. Open Command Prompt
   Press Win + R, type cmd, and press Enter.
2. Ping Localhost
   Type: ping 127.0.0.1
   Checks if your system's network stack is working.

3. **Ping Your Own IP Address**

   Type: ping <your IP>

   Verifies your network adapter is functioning.

4. **Ping Default Gateway**

   Type: ping <gateway IP> (e.g., 192.168.1.1)

   Checks connection to router.

5. **Ping External Website**

   Type: ping google.com

   Tests internet connectivity and DNS.

   Result Interpretation:

- **Reply received → Connection is working**
- **Request timed out → Network issue or blockage**
- **Unknown host → DNS problem**

Section :5

10. Discuss the importance of regular network maintenance and the key tasks involved in maintaining network infrastructure.

Regular network maintenance is important to keep a network secure, fast, and reliable. It helps protect the system from cyberattacks by applying updates and security patches. Maintenance also improves performance by fixing errors and optimizing network settings.
It helps prevent data loss through regular backups and ensures that any problems are detected early before they become serious. Key tasks include updating software, monitoring network activity, checking hardware devices, and troubleshooting issues.

In conclusion, regular maintenance ensures smooth operation, better security, and long-term efficiency of the network.